MTH203: Sequences and series
Lecture 12

1. Some basic observations

We have the following very easy necessary condition for convergence of series.

Theorem 1. *Let $\sum_{n=1}^{\infty}$ be a convergent series. Then $\lim_{n\to\infty} a_n = 0$.*

Proof. Let $s_n := \sum_{i=1}^{n} a_i$ denote the n-th partial sum of the series. Let $\lim_{n\to\infty} s_n = s$.

Let $\epsilon > 0$ be given. Since $\lim_{n\to\infty} s_n = s$, there exists an integer N such that if $n > N$, $|s - s_n| < \epsilon/2$. We will show that if $n > N + 1$, then $|a_n - 0| < \epsilon$.

If $n > N + 1$, then we also have $n > N$ and so $|s - s_n| < \epsilon/2$. Also, $n - 1 > N$ and so $|s - s_{n-1}| < \epsilon/2$. By the triangle inequality,

$$|s_n - s_{n-1}| < |s_n - s| + |s - s_{n-1}| < \epsilon/2 + \epsilon/2 = \epsilon.$$

But $s_n - s_{n-1} = a_n$. Thus, we have shown that $|a_n| < \epsilon$ as required. This shows that $\lim_{n\to\infty} a_n = 0$. □

It is important to realize that this is only a *necessary* condition for convergence. It is not a *sufficient* condition. Indeed, we say in the last lecture that the series $\lim_{n\to\infty}(1/n)$ diverges, despite the fact that $\lim_{n\to\infty}(1/n) = 0$.

Another very easy observation is that to test whether a series converges or not, we do not really have to worry about the initial terms of the series. It is the "tail" that matters.

Proposition 2. *Let $\sum_{n=1}^{\infty} a_n$ be an infinite series in $\mathbb{R}$. Let k be any positive integer. Then, the series $\sum_{n=1}^{\infty} a_n$ converges if and only if the series $\sum_{n=k}^{\infty} a_n$ converges.*

Proof. Let $s_n := \sum_{i=1}^{n} a_i$ be the partial sums of the first series. For any positive integer $k \geqslant n$, we will denote by t_n the partial sum $\sum_{i=k}^{n} a_i$ of the second series. Then, for any integer $n \geqslant k$, we have

$$a_1 + \ldots + a_n = (a_1 + \ldots + a_k) + (a_k + \ldots + a_n).$$

In other words, $s_n = s_k + t_n$ for any $n \geqslant k$.

Now, suppose the first series converges to s. We will show that the second series converges to $s - s_k$. Indeed, for any $\epsilon > 0$, there exists N such that for any $n > N$, we have $|s - s_n| < \epsilon$. Then, if $n > \max\{N, k\}$, we have

$$|s - s_k - t_n| = |s - (s_k + t_n)| = |s - s_n| < \epsilon.$$

Thus, the series $\sum_{n=k}^{\infty} a_n$ also converges.

Conversely, suppose that $\sum_{n=k}^{\infty} a_n$ converges. Let $\lim_{n\to\infty} t_n = t$. We will show that $\lim_{n\to\infty} s_n = t + s_k$. This follows immediately from the following calculation (for $n \geqslant k$):

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty}(s_k + t_n) = \lim_{n\to\infty} s_k + \lim_{n\to\infty} t_n = s_k + t$$

Thus, we see that the series $\sum_{n=1}^{\infty} a_n$ is convergent. □

Practically, this means that when we try to determine whether a series converges or not, we may choose to ignore finitely many initial terms if they happen to be "inconvenient" in some way. We will see an application of this shortly.

2. Geometric series

Let us look at another basic example of a convergent series. First we need a preliminary result:

Proposition 3. *Let x be a real number such that $x > 1$. Then, for any real number y, there exists a positive integer n such that $x^n > y$.*

Proof. First observe that since $x > 1$, $x^n > 1$ for all n. (This can be proved by induction on n.) In particular, $x^n > 0$. Thus, if we multiply the inequality $x > 1$ by x^n, we get $x^{n+1} > x^n$. In other words, the sequence $(x^n)_{n\in\mathbb{N}}$ is strictly increasing.

Now, suppose there exists a real number y such that $x^n \leqslant y$ for any positive integer n. Then the strictly increasing set $\{x^n\}_{n\in\mathbb{N}}$ is bounded above and thus has a supremum. We define $\alpha := \sup\{x^n\}_{n\in\mathbb{N}}$. Since $\alpha > x$, we also have $\alpha > 0$.

Since $x > 1$, we also have $1/x < 1$. (Indeed, if we had $1/x \geqslant 1$, multiplying this inequality by x would give us $1 \geqslant x$, which is a contradiction.) Thus, we have $\alpha/x < \alpha$. Thus, $\alpha - \alpha/x > 0$. Let us set $\epsilon := \alpha - \alpha/x$.

Since $\alpha = \sup\{x^n\}_{n\in\mathbb{N}}$, there exists an integer N such that $x^N > \alpha - \epsilon = \alpha - (\alpha - \alpha/x) = \alpha/x$. Multiplying the inequality $x^N > \alpha/x$ by x, we get $x^{N+1} > \alpha$. This contradicts the fact that α is the supremum of the set $\{x^n\}_{n\in\mathbb{N}}$. Thus, we see that the set $\{x^n\}_{n\in\mathbb{N}}$ cannot be bounded above and thus the real number y with the stated property cannot exist. □

Corollary 4. *Let x be a real number such that $|x| < 1$. Then $\lim_{n\to\infty} x^n = 0$.*

Proof. The result is obviously true if $x = 0$. Thus, for the rest of the proof, we will assume that $x \neq 0$. In particular, this means that $|x| > 0$.

Multiplying the inequality $|x| < 1$ by $|x^n|$, we get $|x^{n+1}| < x^n$. Thus, the sequence $(|x^n|)_{n\in\mathbb{N}}$ is a strictly decreasing sequence.

Let $\epsilon > 0$ be given. Since $|x| < 1$, we have $|1/x| > 1$. (The proof for this statement is similar to an argument in the proof of the above proposition. If we had $|1/x| \leqslant 1$, then multiplying this inequality by the positive number $|x|$, we get $1 \leqslant |x|$, which is a contradiction.)

By the above proposition, there exists a positive integer N such that $|1/x|^N > 1/\epsilon$. Thus, $1/|x|^N > 1/\epsilon$. Multiplying this inequality by the positive number $|x|^N\epsilon$, we get $\epsilon > |x|^N = |x^N|$. Since the sequence $(|x^n|)_{n\in\mathbb{N}}$ is a strictly decreasing sequence, we see that for any $n > N$, we have $|x^n| < |x^N| < \epsilon$.

Thus, we see that for any $n > N$, we have $|x^n - 0| < \epsilon$. Thus, $\lim_{n\to\infty} x^n = 0$. □

Theorem 5. *Let x be a real number. Then the series $\sum_{n=0}^{\infty} x^n$ converges if and only if $|x| < 1$. If $|x| < 1$, it converges to $1/(1-x)$.*

Proof. If $|x| \geqslant 1$, then $|x^n| \geqslant 1$. Thus, by Theorem theorem converging series terms shrink, the series $\sum_{n=0}^{\infty} x^n$ cannot converge. Thus, we now assume that $|x| < 1$.

Let s_n denote the n-th partial sum of the series. Thus,

$$s_n = 1 + x + \cdots + x^{n-1}.$$

Then, we see that

$$xs_n = x + x^2 + \cdots + x^n = (x + \cdots + x^{n-1}) + x^n = (s_n - 1) + x^n.$$

Thus, $(1-x)s_n = (1-x^n)$. (Notice that, so far, we have not used the hypothesis that $|x| < 1$.)

Since $x \neq 1$, we can multiply this equation by $(1-x)^{-1}$ to get $s_n = (1-x^n)/(1-x)$. By Corollary 4, we have $\lim_{n\to\infty} x^n = 0$, and so

$$\lim_{n\to\infty} 1 - x^n = \lim_{n\to\infty} 1 - \lim_{n\to\infty} x^n = 1 - 0 = 1.$$

Thus,

$$\lim_{n\to\infty}(1-x^n)/(1-x) = \frac{\lim_{n\to\infty} 1-x^n}{\lim_{n\to\infty}(1-x)} = \frac{1}{1-x}.$$

□

3. Comparison test for convergence of series

Theorem 6 (Comparison Test)**.** *Let* $(a_n)_{n\in\mathbb{N}}$ *and* $(b_n)_{n\in\mathbb{N}}$ *be sequences of real numbers such that* $0 \leqslant a_n \leqslant b_n$.

(a) If $\sum_{n=1}^{\infty} b_n$*, converges, so does* $\sum_{n=1}^{\infty} a_n$.
(b) If $\sum_{n=1}^{\infty} a_n$ *diverges, so does* $\sum_{n=1}^{\infty} b_n$.

Proof. Let s_n and t_n denote the n-th partial sums of the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$. Note that since $0 \leqslant a_n$, we have $s_{n+1} = s_n + a_{n+1} \geqslant s_n$. Thus, $(s_n)_{n\in\mathbb{N}}$ is a non-decreasing sequence. Similarly, since $0 \leqslant b_n$, the sequence $(t_n)_{n\in\mathbb{N}}$ is also non-decreasing.

First we will prove (a). So, suppose that $\sum_{n=1}^{\infty} b_n$ converges. This means that the sequence $(t_n)_{n\in\mathbb{N}}$ converges. Thus, it is a bounded sequence. Since it is a non-decreasing sequence, it must converge to its supremum. Thus $\lim_{n\to\infty} t_n \geqslant t_m$ for all integers m.

Since $a_n \leqslant b_n$ for all integers n, we may conclude that

$$a_1 + a_2 + \cdots + a_m \leqslant b_1 + b_2 + \cdots + b_m$$

for all integers m. Thus, $s_m \leqslant t_m$ for all m. Since $\lim_{n\to\infty} t_n \geqslant t_m$ for all m, we also have $\lim_{n\to\infty} t_n \geqslant s_m$ for all m. Thus, the non-decreasing sequence $(s_n)_{n\in\mathbb{N}}$ is bounded above. Thus, the set $\{s_n\}_{n\in\mathbb{N}}$ has a supremum and the sequence $(s_n)_{n\in\mathbb{N}}$ must converge to this supremum. This completes the proof of (a).

Now, let us prove (b). As we noted above, the sequence $(s_n)_{n\in\mathbb{N}}$ is non-decreasing. Thus, if it does not converge, it cannot be bounded. (This is because any *monotone* bounded sequence is necessarily convergent. Remember that a bounded sequence may not be convergent unless we impose some additional hypothesis.) Thus, for any real number x, there exists n such that $s_n > x$. Since $s_n \leqslant t_n$, we also have $t_n > x$. This shows that the sequence $(t_n)_{n\in\mathbb{N}}$ is also not bounded. Thus, it cannot be convergent. □

Remark 7. Note that the assumption that $0 \leqslant a_n$ is very important for part (a). For instance, consider the series $\sum_{n=1}^{\infty}(-1)/n$ and $\sum_{n=1}^{\infty}(1/2)^n$. Then $(-1)/n < (1/2)^n$ for all n. However, the second series is convergent and the first one is not.

Example 8. We will now show that the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. This is done by the comparing this series to one with slightly larger terms, but which is easily summable.

For every $n \geqslant 2$, we have $\frac{1}{n^2} < \frac{1}{n(n-1)}$. We will show that the series $\sum_{n=2}^{\infty} \frac{1}{n(n-1)}$ converges. This is done by the following rather clever trick:

$$\frac{1}{n(n-1)} = \frac{n-(n-1)}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$$

Thus, the partial sums of this series are calculated as follows:

$$t_m := \sum_{n=2}^{m} \frac{1}{n(n-1)} = \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{m-1} - \frac{1}{m}\right)$$

It is easy to see that the second term within each pair of parentheses cancels out the first term within the next pair of parentheses. (This is an example of what is called a *telescopic sum.*) Thus, we get $t_m = 1 - \frac{1}{m}$. Thus, we see that $t_m < 1$ for all m. The sequence $(t_m)_{m\geqslant 2}$ is clearly strictly increasing. Since it is bounded above, it is necessarily convergent. We also see that the limit is less than or equal to 1.

Thus, by the Comparison Test, we see that the series $\sum_{n=2}^{\infty} \frac{1}{n^2}$ is convergent. By Proposition proposition tail of series, this implies that the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

4. Reading suggestions

The material in this lecture corresponds to parts of Sections 2.4 and 2.5 of the textbook ([1]).

References

[1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015.